

Weight — Final Tool Revision

Revisions driven by the usability evaluation. Everyday Tools — Fordham Wong

Revised

tool fwong29.github.io/reusable-studio-engine/everyday-tools/

Grapher </everyday-tools/grapher/> (new — graphing, derivatives, integrals)

Repository github.com/fwong29/reusable-studio-engine

1 · KEY USABILITY FINDINGS

Tester: Will — a teen who does a lot of math regularly and is highly comfortable with technology. He was given only the deployed link and told how to open it, then left to use and explore the tool on his own with no explanation of the interface. This fits the intended audience: a fast, fluent user for whom entering numbers is automatic, so his reactions are a fair read on whether the friction lands or just annoys.

Major issues observed:

- **Controls weren't self-explanatory.** He said the function of each button wasn't clear and explicitly asked for clearer instructions — the keyboard mapping (Shift, space) was the sharpest point of confusion.
- **The tool felt limited.** He wanted it to "do more math things"; with only $+$, $-$, $\times$, $\div$ there wasn't much for the friction to be about.
- **Friction read as inefficiency, not intent.** He rated it 3/5, called the intervention "annoying, too inefficient," and said he wouldn't use it — but grasped the purpose ("be mindful of my calculations") and liked the numbers and overall concept. The signal: the deliberate cost is fine; what hurt was not knowing how to drive the interface.

2 · REVISION LOG

Revision 1 — Clearer instructions

Issue Identified: Users couldn't tell what each control did.

Feedback Evidence: "The function of each button wasn't clear"; "make the instructions more clear"; confusion at "the key combos like shift + space."

Design Change Implemented: Added an always-visible legend naming the number key, the operator key, and `= /` `C / <`, plus an expandable "keyboard shortcuts" list (space = one tap, Shift = cycle operator, = / Enter = solve, Backspace = undo, Esc = clear).

Revision 2 — More math

Issue Identified: The tool did too little to feel worth the friction.

Feedback Evidence: "Make it so that I can do more math things."

Design Change Implemented: Grew the tool into a graphing calculator on one page — calculator left, live graph right. Plain arithmetic still shows a number ($5 \times 2 = 10$); type an expression in `x` and it graphs. A cycling key handles the four basic operators `+ - × ÷`, with `^` and `%`, the variable `x`, parentheses, functions (`sin cos tan ln log`), and an **nth-root** (index typed into a box) as their own keys. Above the graph, mutually-exclusive **d/dx** and **∫** toggles drive the analysis, and the answer lands in a box under the display: d/dx shows the exact symbolic **derivative and indefinite integral** ($\int y \, dx = \dots + C$), while **∫** shows a numeric **definite** integral whose limits are **click-to-set boxes**. Hovering the curve reads out coordinates, and +/- buttons zoom. Crucially the one rule holds everywhere — every number, including a root index or an integral limit, is still multi-tapped.

Revision 3 — Atmosphere / responsiveness

Issue Identified: The interface felt inert, which made the friction read as the tool being slow rather than deliberate.

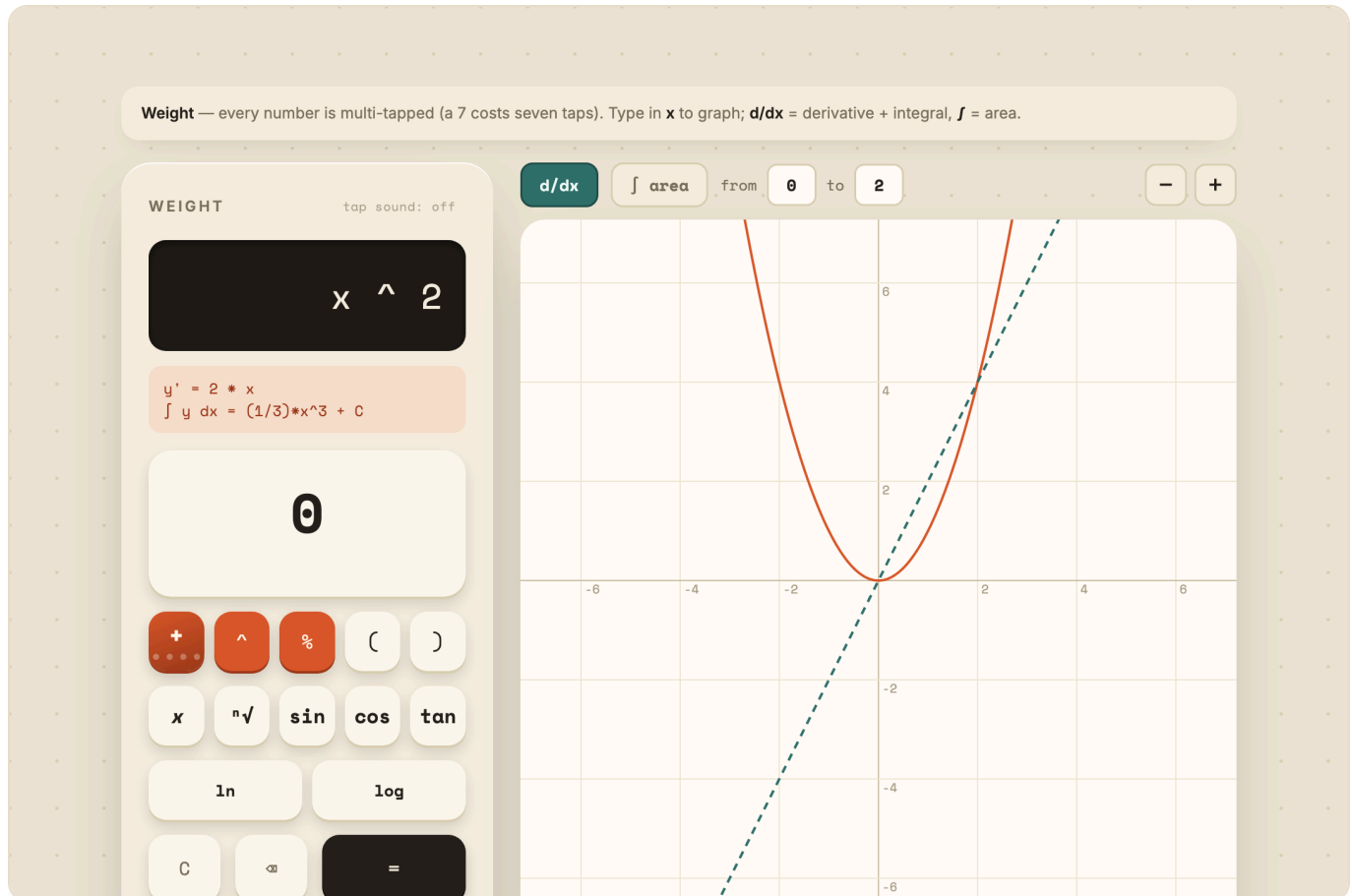
Feedback Evidence: Intervention felt "too inefficient"; the surface gave nothing back as he used it.

Design Change Implemented: Added an ambient background field (inspired by the Netflix Drop page) — a grid of dots that pushes away from the cursor while the area around it darkens — so the tool feels alive and responsive to presence without touching the deliberate cost of a tap. Respects `prefers-reduced-motion`.

3 · UPDATED TOOL

Live: fwong29.github.io/reusable-studio-engine/everyday-tools/

Revised Weight — one page: multi-tapped calculator left (with legend, cycling keys and their position pips, nth-root and parentheses), live graph right showing x^2 and its derivative.



4 · REFLECTION

Watching Will use it taught me something I could not have seen from inside the design: I had built the friction so carefully that I stopped noticing you first have to understand the controls before the friction can even mean anything. To me the number key and the operator key were obvious, so I never questioned them — but he stalled on "shift + space" and read the whole tool as "inefficient," when the real problem was that he didn't know how to drive it. The deliberate cost was never the issue; the missing map was. A real user can't tell intentional friction from an unclear interface, and only by observing one silently did I learn that my job wasn't to defend the friction but to make the controls legible enough that the pause could be felt as a choice instead of a wall.